

OKNexus Exchange

Where the build is today, and the architecture ahead

Prepared for Mr Sam Soair · 14 July 2026 · Development update

This note confirms we are aligned on direction and lays out, in plain terms, what has already been built, what changes now that the provider stack is confirmed, and how the custodial wallet system will work. The short version: **we are building the right product** — a custodial exchange running on enterprise infrastructure — and the platform was deliberately built so that the named providers slot into seams that already exist, rather than requiring a rebuild.

1. The model we are building (confirmed)

OKNexus is a **custodial exchange built on enterprise-grade third-party infrastructure** — not a self-built blockchain, not a crypto wallet, and not a Binance dashboard. A licensed custody provider secures customer funds and keys; OKNexus owns the product, the experience, and the business logic on top. Our responsibility is the premium frontend, the backend logic, the API integrations, the admin and compliance portals, automation, notifications, reporting, and platform intelligence.

2. What is already built and live

A working platform is deployed and running at oknexusexchange.com. The following are complete and reusable as-is — they are the product layer that sits above the providers:

- **Premium frontend & design system** — marketing site plus the full application UI.
- **Accounts & sessions** — registration, login, roles, suspension.
- **Balance ledger** — a double-entry accounting system that tracks exactly what each user owns (the heart of any custodial exchange).
- **Wallet screens** — deposit, withdraw, balances, and transaction history, with the full pending → confirmed → failed lifecycle.
- **Trading** — Spot (order book, market & limit orders, charts, history), Instant Swap, Buy/Sell (ramp), OTC desk.
- **P2P marketplace** — post buy/sell ads, escrow-protected trades, in-trade chat, disputes.
- **Admin portal** — users, disputes, transactions, KYC review, analytics, ad moderation.
- **KYC intake** — submission flow plus admin approval workflow.

3. The enterprise provider stack

Each "engine" below is currently running as a working prototype we built to prove the flows. Going to production means **swapping each prototype for the named provider**, behind interfaces that are already in place.

Area	Provider	Today	Action
Custody & wallets	Fireblocks (or Alchemy)	Self-built test wallets	Integrate
Identity / KYC	Sumsub or Veriff	Manual admin review	Integrate

Area	Provider	Today	Action
AML & sanctions	Chainalysis	—	Integrate
Spot market & matching	Liquidity provider	Live prices + demo matching	Integrate
Instant swap	Swap API	Internal pricing	Integrate
Fiat on/off-ramp	MoonPay or Bitnob	Simulated	Integrate

4. How the custodial wallet will work

Think of it as two layers working together — **the vault** and **the passbook**.

- **The vault (Fireblocks)** holds the actual cryptocurrency and the private keys. Keys are never held on our servers; Fireblocks secures them and handles all on-chain signing, plus the hot / warm / cold separation that protects the bulk of funds. This is the licensed, insured part.
- **The passbook (our ledger)** is the record of who owns what. A customer's balance in the app is their ledger balance — their share of the pooled funds in the vault. This layer is already built.

Deposits

- 1 The customer opens "Deposit," picks an asset, and sees a unique address (and QR code) that Fireblocks assigns to them.
- 2 They send crypto to that address from anywhere.
- 3 Fireblocks detects the incoming funds and notifies our backend as confirmations build up.
- 4 Once confirmed, we credit the customer's ledger balance — and it appears in their account.

Withdrawals

- 1 The customer requests a withdrawal to an external address.
- 2 We check and reserve their ledger balance, screen the destination (Chainalysis), and require 2FA.
- 3 We instruct Fireblocks to send; its policy engine auto-approves small amounts and escalates large ones for extra approval.
- 4 When the transaction confirms, we finalize the deduction; if it fails, the reserved funds are released automatically.

Trading, swaps, and internal transfers

These never touch the blockchain. When two users trade, or a swap or order fills, only the passbook entries move — instantly, with no network fees. Crypto only moves on-chain at the edges: real deposits coming in and real withdrawals going out. This is exactly why a custodial exchange can offer instant, free internal trading.

Why this is the safe path. The alternative — holding customers' private keys on our own servers — carries security and licensing risk that is not appropriate for real customer funds. Fireblocks is purpose-built for this and already secures a large share of institutional crypto. We integrate it; we do not rebuild it.

5. What we keep vs. what we swap

The large majority of the platform stays. The customer-facing wallet screens and the accounting behind them do not change — in production they are simply driven by Fireblocks instead of our test wallets. What gets replaced is the behind-the-scenes engine, not the product.

- **Keeps:** all UI, the balance ledger, wallet & transaction screens, trading screens, admin portal, P2P, accounts, the design system.
- **Swaps in:** Fireblocks custody, Sumsb/Chainalysis compliance, liquidity-provider spot, Swap API, MoonPay/Bitnob ramp.

6. Still to build for launch

- **Account security** — email & phone verification, Google Authenticator (2FA), password reset, device management, login history.
- **Trading** — stop orders (market & limit are done).
- **Wallet extras** — QR codes, clearer pending/failed states, fee display, internal transfers between users.
- **Platform services** — notifications, reporting, and admin/compliance analytics.
- **Later phases** — earn/staking products and the native utility token from the original agreement.

7. What we need from you

The integrations can be built now against each provider's sandbox. Going live requires the provider accounts to exist in the company's name. To keep things moving, please confirm:

- Which provider accounts already exist or are in progress (Fireblocks, Sumsb/Veriff, Chainalysis, MoonPay/Bitnob, the liquidity provider, the swap API).
- Most providers require company onboarding / business verification and API keys; custody in particular is an enterprise arrangement. Knowing the budget and licensing position for Fireblocks lets us confirm it as the custody choice or line up an interim alternative (e.g. Alchemy).

8. Suggested build order

- 1 **Fireblocks custody** — unlocks real deposits and withdrawals; the foundation everything else settles onto.
- 2 **Account security (2FA & verification)** — required for a real exchange, and independent of the providers so it can run in parallel.
- 3 **KYC + AML (Sumsb + Chainalysis)** — the compliance gate before real money moves.
- 4 **Ramp + Swap (MoonPay/Bitnob + Swap API)** — the money-in and convert paths.
- 5 **Liquidity-provider spot + stop orders** — full professional trading.

OKNexus Exchange — internal development update. Provider names are the client's confirmed selections and are subject to commercial onboarding. Nothing in this document is financial or legal advice.